

AWS S3 Cross Region Replication: -

Summary: -

In this lab, I configured AWS S3 Cross Region Replication (CRR) by creating a source bucket in Mumbai and a destination bucket in the US East region. I enabled bucket versioning, created a replication rule, selected the destination bucket, and configured IAM permissions for replication. After saving the configuration, all files from the source bucket were automatically replicated to the destination bucket in another AWS region successfully

Let's Start: -

- CRR automatically replicates every object uploaded to an S3 bucket to a destination bucket in a different AWS Region.
- Basically, I need to create two S3 Bucket (source Bucket and destination Bucket) in two different regions.

General purpose buckets (1/199) [Info](#)

 [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

[×](#) 1 match < 1 > [⚙️](#)

Name	AWS Region	Creation date
s3destination2312	US East (N. Virginia) us-east-1	May 22, 2026, 15:21:05 (UTC+05:30)

- This is destination bucket present in US East (N. Virginia).

General purpose buckets (1/199) [Info](#)

 [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

[×](#) 1 match < 1 > [⚙️](#)

Name	AWS Region	Creation date
s3source1223	Asia Pacific (Mumbai) ap-south-1	May 22, 2026, 15:07:00 (UTC+05:30)

- This is source bucket present in Asia Pacific (Mumbai).
- Now open source bucket.

Objects (2)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open ↗](#) [Delete](#) [Actions ▼](#) [Create folder](#) [Upload ↗](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

< 1 > ⚙

☐	Name	Type	Last modified	Size	Storage class
☐	EBS Volume.pdf	pdf	May 22, 2026, 15:07:37 (UTC+05:30)	1.7 MB	Standard
☐	Elastic File Storage.pdf	pdf	May 22, 2026, 15:07:38 (UTC+05:30)	718.2 KB	Standard

- You see I added two files in Source bucket.
- Now I need to do our document transfer to destination Bucket that present in another region.

s3source1223

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Lifecycle rules

[Refresh](#) [View details](#) [Edit](#) [Delete](#) [Actions ▼](#) [Create lifecycle rule](#)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Lifecycle rule na...	Status	Scope	Current version ...	Noncurrent vers...	Expired object d...	Incomplete mul...
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No lifecycle rules

- In source click to the Management.
- Scroll down.

Replication rules (0)

[Refresh](#) [View details](#) [Edit rule](#) [Delete](#) [Actions ▼](#) [Create replication rule](#)

Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS encryption object KMS DSS
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No replication rules

You don't have any rules in the replication configuration.

[Create replication rule](#)

- Click Create replication rule.

Create replication rule [Info](#)

⚠ Replication requires versioning to be enabled for the source bucket. Enable object versioning on this bucket to continue creating the replication rule.

[Enable Bucket Versioning](#)[Diagnose with Amazon Q](#)

- First replication require versioning Enable simply click Enable Bucket Versioning.

Replication rule configuration

Replication rule name

rule-1

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status

Choose whether the rule will be enabled or disabled when created.

- ☒ Enabled
☐ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

- Enter name of Replication rule.
- Status by default Enable.
- Scroll down.

Source bucket

Source bucket name

s3source1223

Source Region

Asia Pacific (Mumbai) ap-south-1

Choose a rule scope

- ☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

- Now choose a rule scope Apply to all objects in the bucket.
- Scroll down.

Destination

Destination

You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)

- ☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name

Choose the bucket that will receive replicated objects.

Enter bucket name

Browse S3

Destination Region

-

- Now here I have to choose destination.
- Destination can be from your own account or it can be from other account as well.
- If you want to store data into another AWS account, you have to get account ID
- Now I am going with bucket in the same AWS account.
- Click Browse.

Choose a bucket

S3 Buckets

Buckets (1/199)

Search: s3des 1 match

Name	AWS Region
s3destination2312	US East (N. Virginia) us-east-1

Cancel Choose path

- Now search and select destination Bucket and click Choose path.

Bucket name

Choose the bucket that will receive replicated objects.

s3destination2312 [Browse S3](#)

✖ Replication requires Versioning to be enabled for the destination bucket
Enable Versioning on this bucket to continue creating the replication rule or select a different bucket.

[Enable Versioning](#) [Diagnose with Amazon Q](#)

Destination Region

US East (N. Virginia) us-east-1

- Now we need to click Enable Versioning.
- scroll down.

IAM role

Permission to access the specified resources

- ☒ Create new role
- ☐ Choose from existing IAM roles
- ☐ Enter IAM role ARN

- and choose Create new role.
- Scroll down.

Additional replication options

☐ Replication Time Control (RTC)
Replication Time Control replicates 99.99% of new objects within 15 minutes and includes replication metrics. Additional fees will apply. [Learn more](#)

☐ Replication metrics
With replication metrics, you can monitor the total number and size of objects that are pending replication, and the maximum replication time to the destination Region. You can also view and diagnose replication failures. CloudWatch metrics fees apply. [Learn more](#) or see [Amazon CloudWatch pricing](#)

☐ Delete marker replication
Delete markers created by S3 delete operations will be replicated. Delete markers created by lifecycle rules are not replicated. [Learn more](#)

☐ Replica modification sync
Replicate metadata changes made to replicas from the destination bucket to the source bucket. [Learn more](#)

Cancel Save

1. RTC Disabled (Default): - Replication Time Guarantee No guarantee; replication time can vary from minutes to hours.
RTC Enable: - Replication Time Guarantee Provides a predictable replication time of 99.99% objects within 15 minutes.

2. Replication metrics: - Provides detailed insights into the status and performance of your replication process.
3. Delete Marker Replication Disable (Default): - Preserves objects in the destination bucket even if they're deleted from the source.
Delete Marker Replication Enabled: - When you delete an object in the source, it's also deleted in the destination.
4. Replica modification sync: - Ensures changes made to the replicated copies of objects in the destination bucket are synced back to the source bucket.
5. I am not going to allow any of them I click Save.

Replicate existing objects?



You can enable a one-time Batch Operations job from this replication configuration to replicate objects that already exist in the bucket and to synchronize the source and destination buckets. [Learn more](#) or [see pricing](#)

Existing objects

- ☐ No, do not replicate existing objects.
- ☒ Yes, replicate existing objects.

Cancel

Submit

- Now click Yes, then Submit.

Create Batch Operations job

Job settings

A job is used to execute batch operations on a list of S3 objects. The list of objects is contained in a replication manifest object generated by S3.

Job run options

You can choose whether to have the job start automatically after the replication manifest is generated or to have the job wait in the *Awaiting your confirmation to run* status until you run the job.

☒ Automatically run the job when it's ready

When selected, the job automatically runs without waiting for you to start it.

☐ Wait to run the job when it's ready

Recommended if you want to review the manifest or job details before running the job.

Completion report

Generate a CSV completion report that lists your target objects, task success or error codes, outputs, and descriptions. Completion reports are encrypted using SSE-S3. [Learn more](#)

☐ Generate completion report

- Now Off Completion report.
- Scroll down.

Permissions

Choose an IAM role with the [required access permissions and trust relationships](#). An IAM role policy template based on your job configuration, and the IAM trust policy required for batch operations to assume the IAM role are available below. [Learn more about IAM roles](#).

► View IAM role policy template and IAM trust policy

Permission to access the specified resources

☒ Create new role

☐ Choose from existing IAM roles

☐ Enter IAM role ARN

Cancel

Save

- Now Select Create new role and click save.
- Now go to open Aws console in new tab open S3 and go inside destination folder its works or not.

s3destination2312 Info

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [File systems - new](#) | [Access Points](#)

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☐

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☐  Elastic File Storage.pdf	pdf	May 22, 2026, 15:07:38 (UTC+05:30)	718.2 KB	Standard

- Yes, it's working properly.